

ENOCH MOK

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SKILLS

Programming: Python, JavaScript, SQL

Machine Learning & Modeling: scikit-learn, PyTorch, OpenCV, Numpy, pandas

Visualization & Reporting: Tableau, Geospatial Analysis (GeoPandas, Folium), Matplotlib, Plotly

Spoken Languages: English, Mandarin, Cantonese

EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY

Singapore

Bachelor of Engineering - CN Yang Research Scholar

Aug 2020 - May 2024

Computer Science (Specialization: AI & Data Science)

Relevant Coursework: Data Analysis; Software Engineering; Operating Systems

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA, USA

Exchange Programme in Computer Science

Jul 2022 - Dec 2022

Relevant Coursework: Computer Vision; Computer Networking; Artificial Intelligence

WORK EXPERIENCE

SingHealth

Singapore

Data Analyst

Dec 2024 - Ongoing

- Pioneered scalable, automated ETL data pipelines and analytics dashboards (Python, pandas, SQL, Tableau) to visualize metrics from 1.5M+ patient records, resulting **\$18M+** in cost savings.
- Spearheaded geospatial analytics with custom geocoding, spatial joins, and multi-level geographic aggregation to support regional facility planning, driving enrolment up by **11%** (437K to 484K).
- Consolidated 5 Excel processes into an integrated Tableau analysis platform adopted by 30+ senior health leaders for strategic planning, reducing manual reporting effort by 10+ hours per week.
- Awarded SingHealth "You Shine" Award as part of a cross functional team that standardised interactive dashboards visualising community engagement, population health and demographic data.

Augmented Intelligence and Multimodal Analytics (AIM) for Health Lab

Melbourne, Australia

Research Assistant at Monash University

Jun 2023 - Dec 2023

FYP Title: Federated Learning for Breast Cancer Classification

- **Published an academic book chapter** on Federated Learning for medical imaging after validating a decentralized training pipeline, achieving **73% recall** on ~32k mammography images.
- Engineered a privacy-preserving distributed training framework (Flower, PyTorch) across 4 simulated healthcare nodes, benchmarking neural network architectures against multi-institutional data constraints.
- Designed robust DICOM preprocessing pipelines (Pydicom, OpenCV) to normalize heterogeneous scanner data, utilizing GradCAM to validate clinical interpretability and mitigate cross-site domain shifts.

Iota Medtech (5-employee MedTech startup)

Singapore

Software Engineer Intern

May 2022 - Aug 2022

- Architected a production-ready Electron.js desktop application for lung cancer screening, achieving **95% diagnostic accuracy** and deploying the solution for clinical use within a 3-month internship.
- Streamlined a high-throughput DICOM conversion and preprocessing pipeline (Pydicom, OpenCV) to feed AI models for X-ray/CT analysis, processing 1,000+ patients in less than 6 minutes.
- Integrated local model inference with backend AI via secure RESTful APIs and encrypted local authentication and data storage (PouchDB), preserving patient privacy.

PROJECTS

HDB-Public Transport System Analysis

- Produced automated Python pipelines to extract HDB and public transport data from multiple RESTful APIs and open data sources using pandas to create datasets for analysis.
- Designed an equity scoring index to quantify underservice and applied predictive modeling to rank and prioritize transport projects over a 20 year horizon.
- Designed geospatial visualizations with GeoPandas and Folium to map transport coverage and population across 40 residential areas and documented findings in reproducible notebooks.

AI-Optimized HIV Treatment in Developing Countries

- Developed a Reinforcement Learning pipeline to optimize personalised Dynamic Treatment Regimes for HIV patients, achieving a **Top-8 ranking** at the MLDA Hackathon.
- Engineered custom OpenAI Gym environments with cohort stratification to evaluate model performance, ensuring clinically meaningful reward and state representations.

INTERESTS

Music: Guitarist; Bassist; Music Director (Band)

Sports: Ultimate Frisbee (Coach/Captain)

IT: Home Server; Personal Website